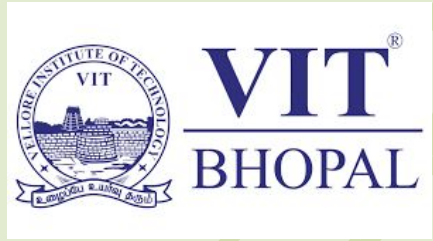
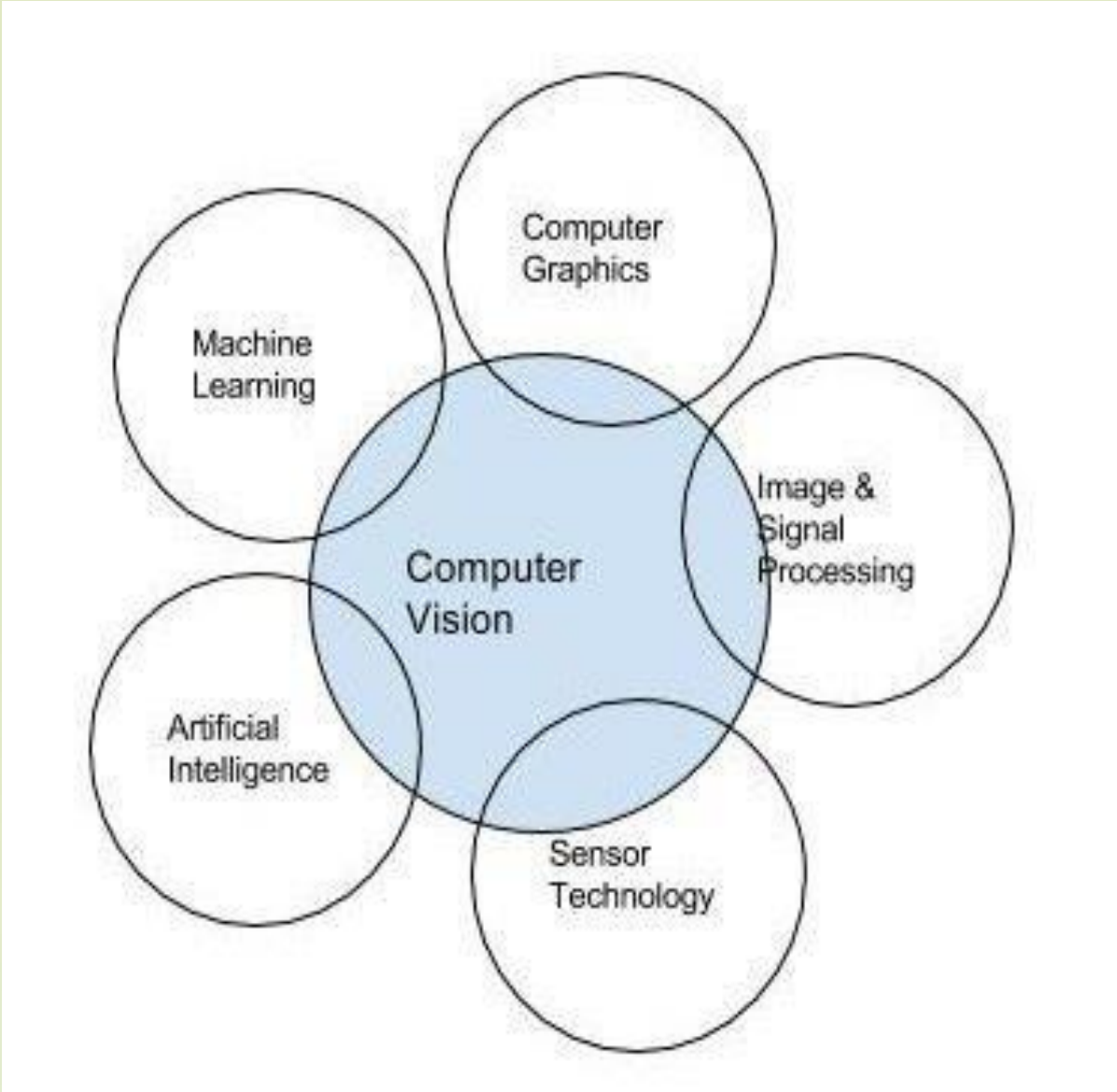
COMPUTER VISION (CSE3010)

Dr. Gaurav Soni
Sr.Assistant Professor
School of Computing Science Engineering



INTRODUCTION TO COMPUTER VISION

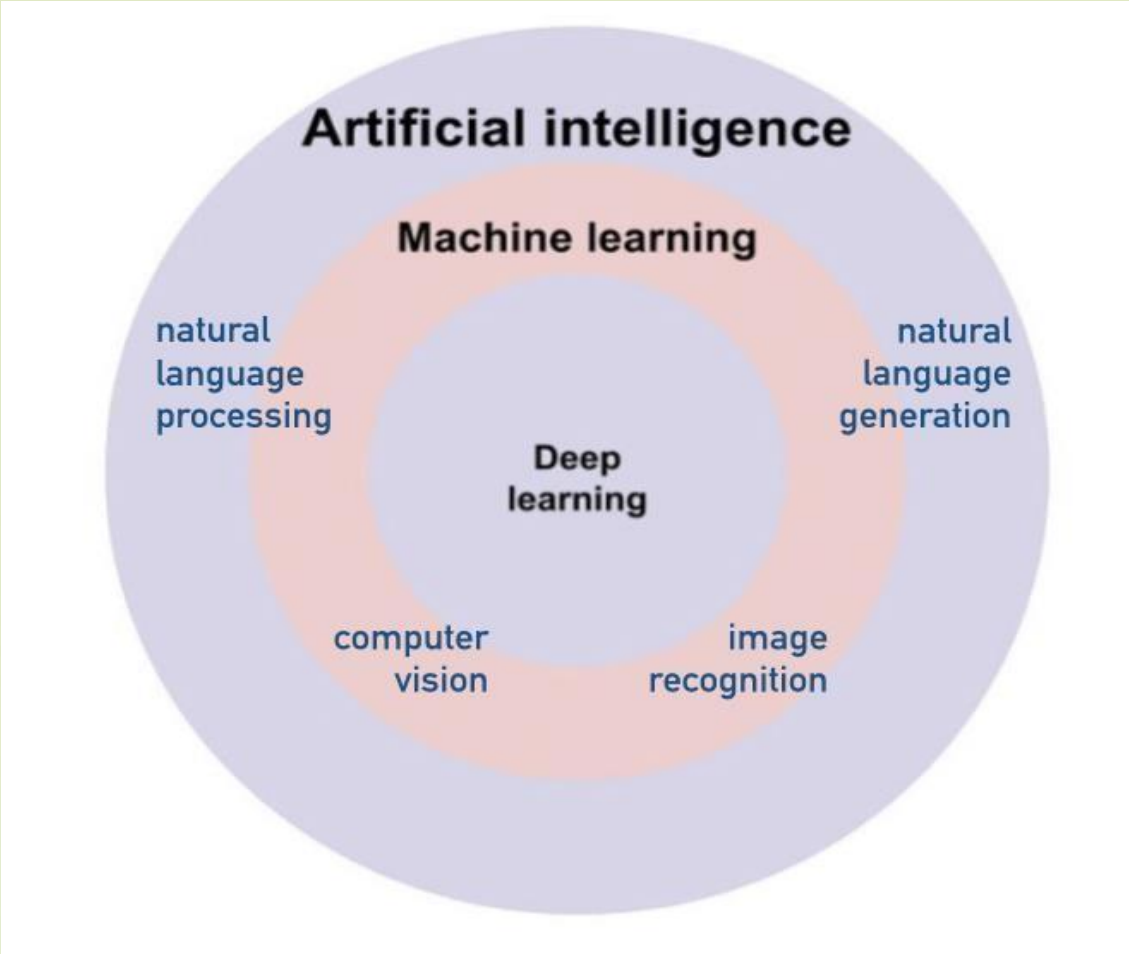
- **Computer vision is a field of artificial intelligence (AI) that trains machines to interpret and understand visual information from the world.**
- **Computer vision** is an interdisciplinary scientific field that deals with how computers can be made to gain high- level understanding from digital images or videos.
- From the perspective of engineering, it seeks to automate tasks that the human visual system can do.



INTRODUCTION TO COMPUTER VISION

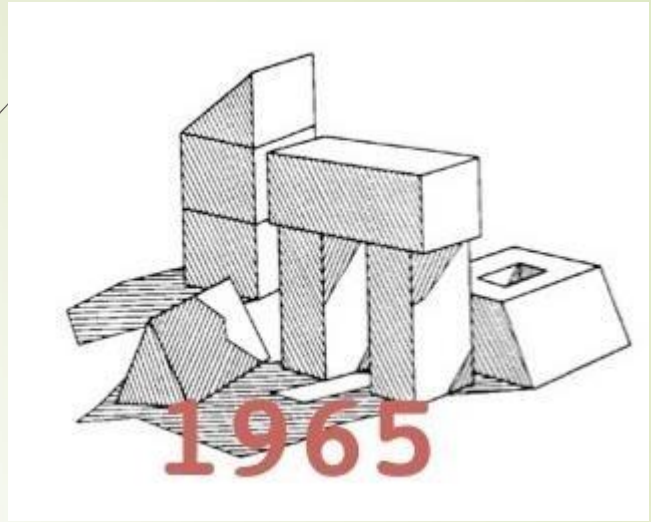
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A brief history of Computer Vision and AI Image Recognition

The early days of computer vision





As computer vision evolved, algorithms started to be programmed to solve individual challenges, and they become better at doing the job the more they repeat the task

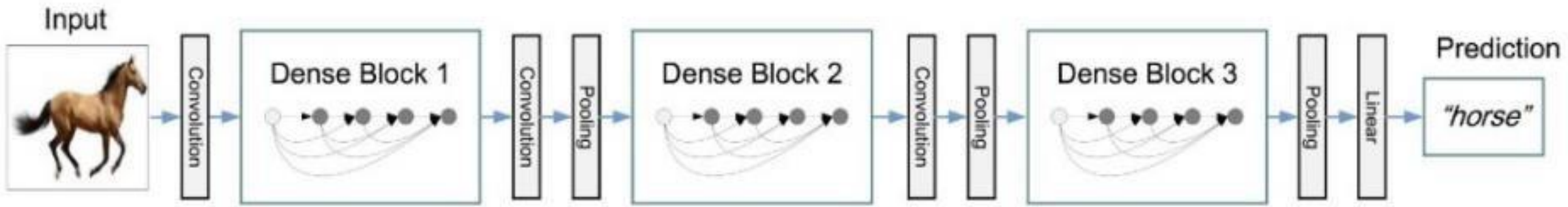
INTRODUCTION TO COMPUTER VISION



Example: CCTV cameras in London Underground stations programmed to spot static / abandoned objects

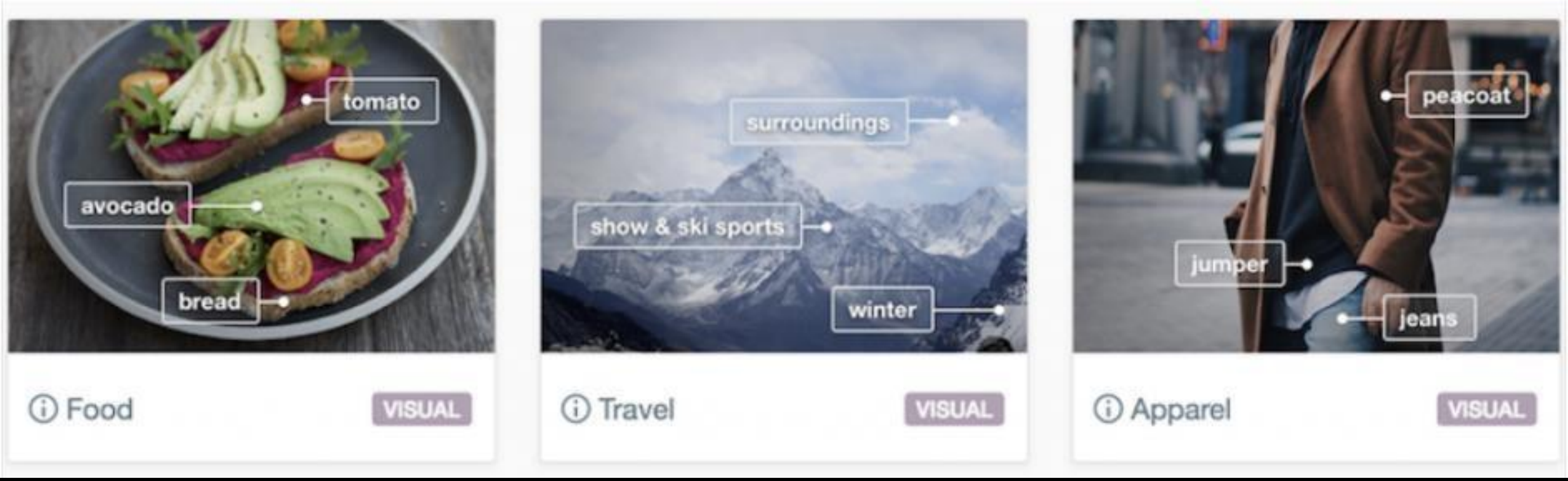
INTRODUCTION TO COMPUTER VISION

Deep learning and Vertical Image Recognition



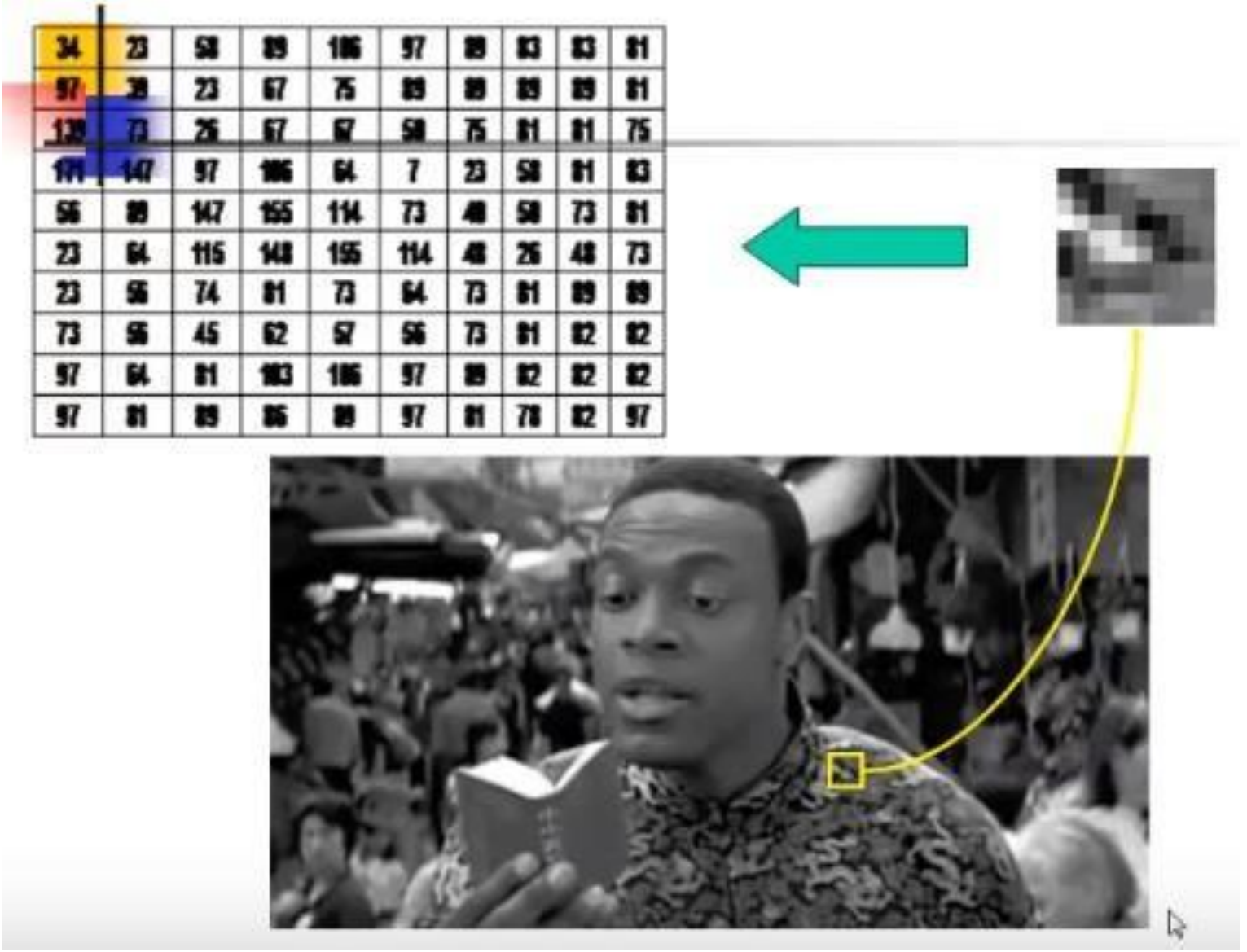
DenseNet visualization via Deep Learning for Image Recognition: why it's challenging, where we've been, and what's next

INTRODUCTION TO COMPUTER VISION



Image

- 2-D array of numbers (intensity values, gray levels)
- Gray levels 0 (black) to 255 (white)
- Color image is 3 2-D arrays of numbers
  - Red
  - Green
  - Blue
- Resolution (number of rows and columns)
  - 128X128
  - 256X256
  - 512X512
  - 640X480


34 23 58 89 186 97 88 83 83 81
97 38 23 67 75 89 88 89 89 81
120 73 25 67 67 58 75 81 81 75
191 147 97 186 64 7 23 58 81 83
56 88 147 155 114 73 48 58 73 81
23 64 115 148 155 114 48 25 48 73
23 56 74 81 73 64 73 81 89 89
73 56 45 62 57 56 73 81 82 82
97 64 81 183 186 97 88 82 82 82
97 81 89 85 88 97 81 78 82 97

Deep Learning for Effective Detection of Excavated Soil Related to Illegal Tunnel Activities

RESEARCH IN COMPUTER VISION



Fig. 1: One tunnel discovered in 2006. Mexico is on the top and U.S. is at the bottom. The white dash line shows the path of the tunnel.



Fig. 2: Soil samples in WV-2.

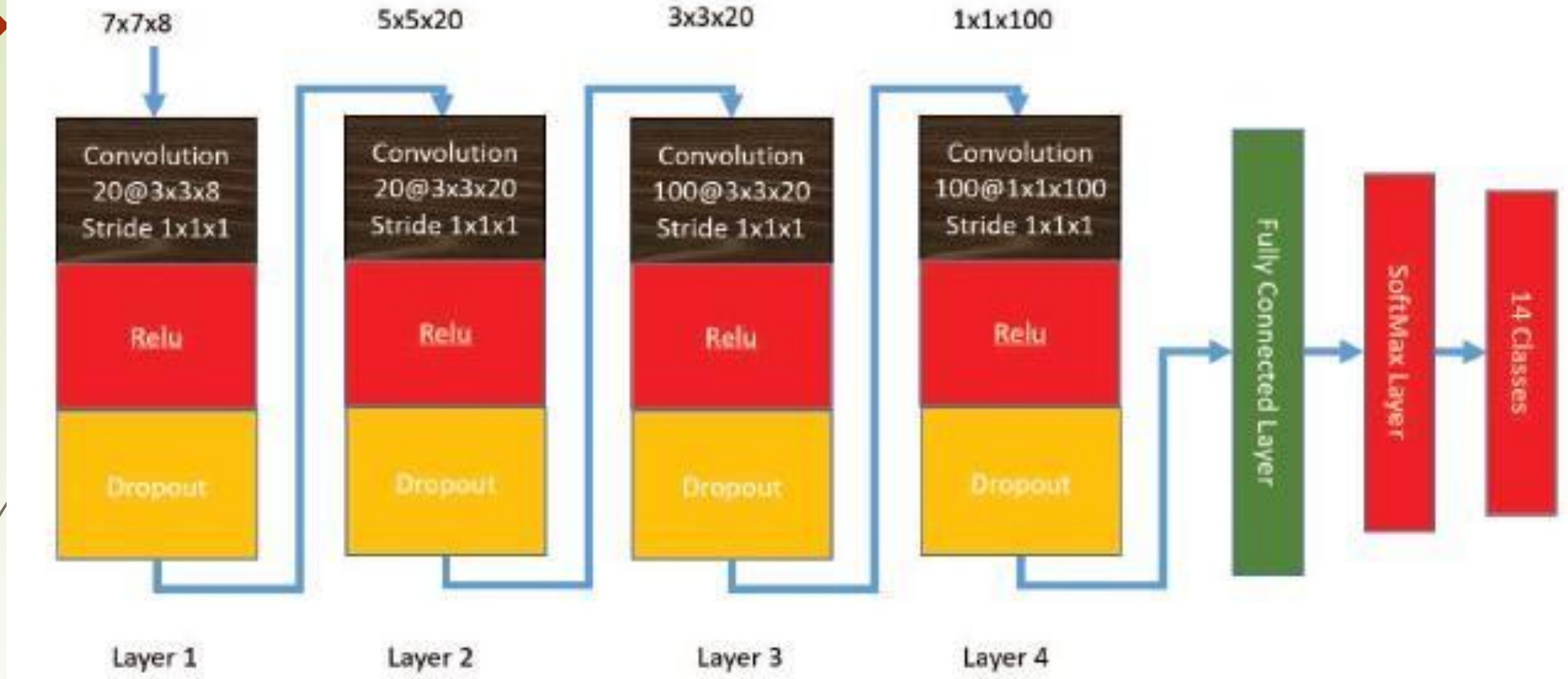


Fig. 3: CNN structure for soil detection on the 8 channel pan-sharpened multispectral images.

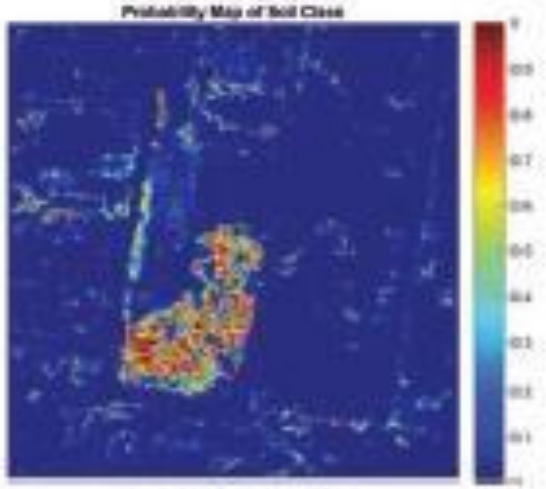


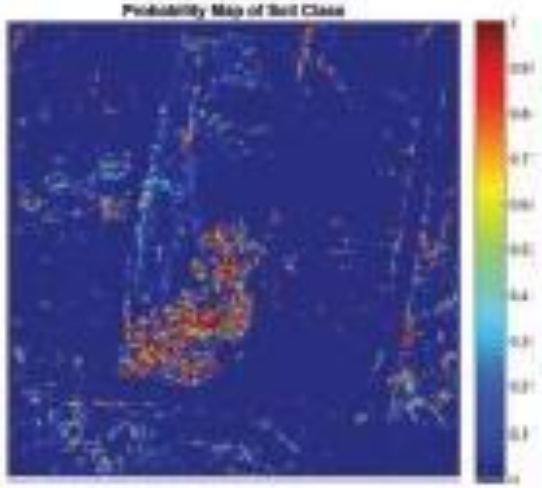
(a) Testing image 1 captured on 03/19/2010

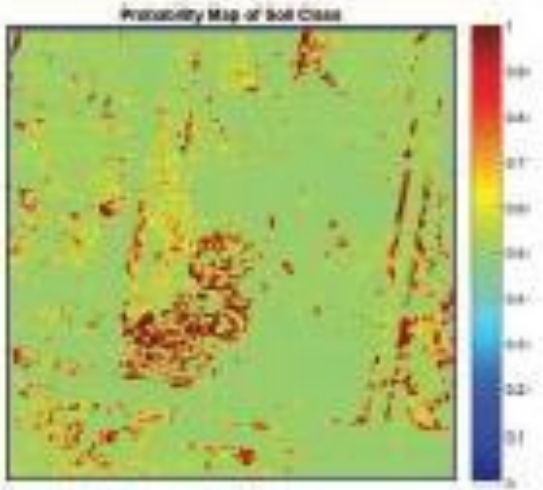


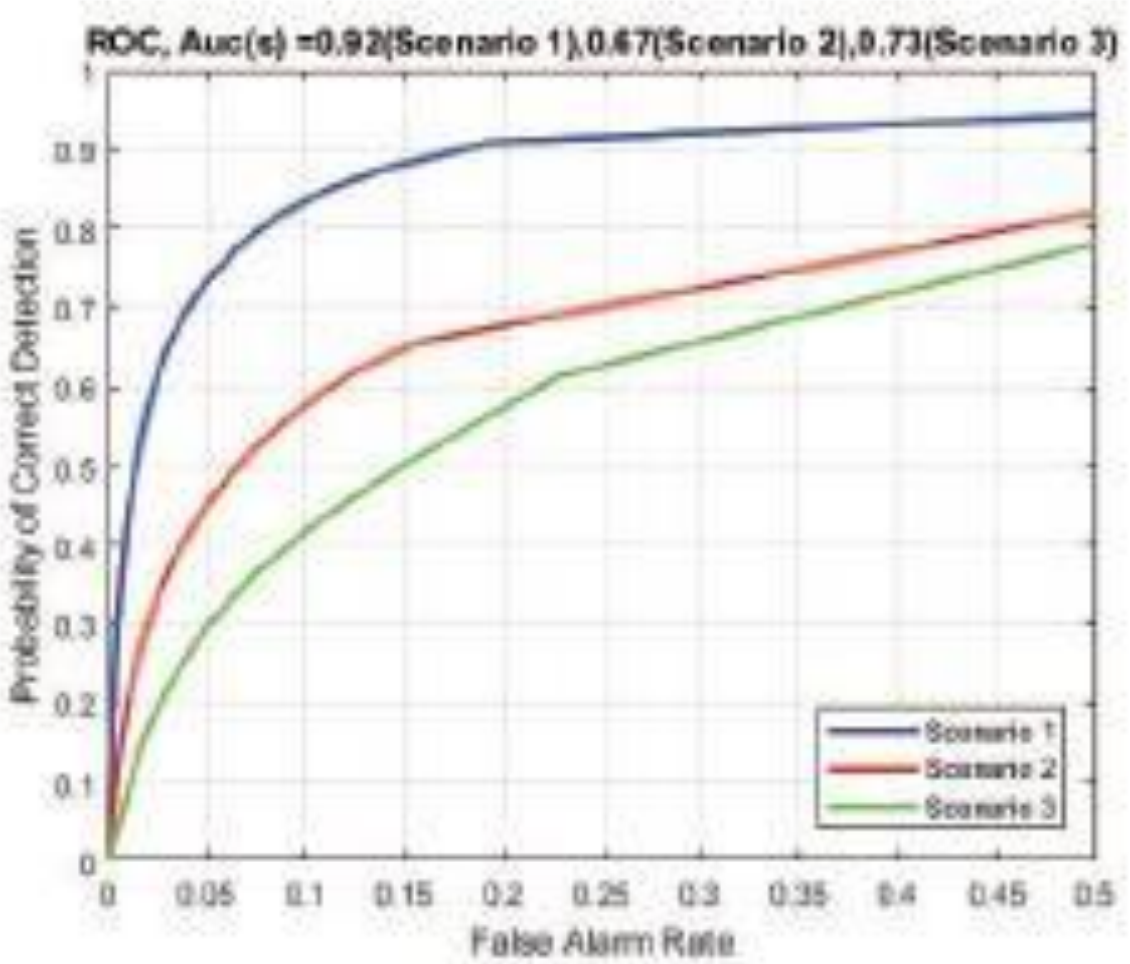
(b) Testing image captured on 10/11/2010



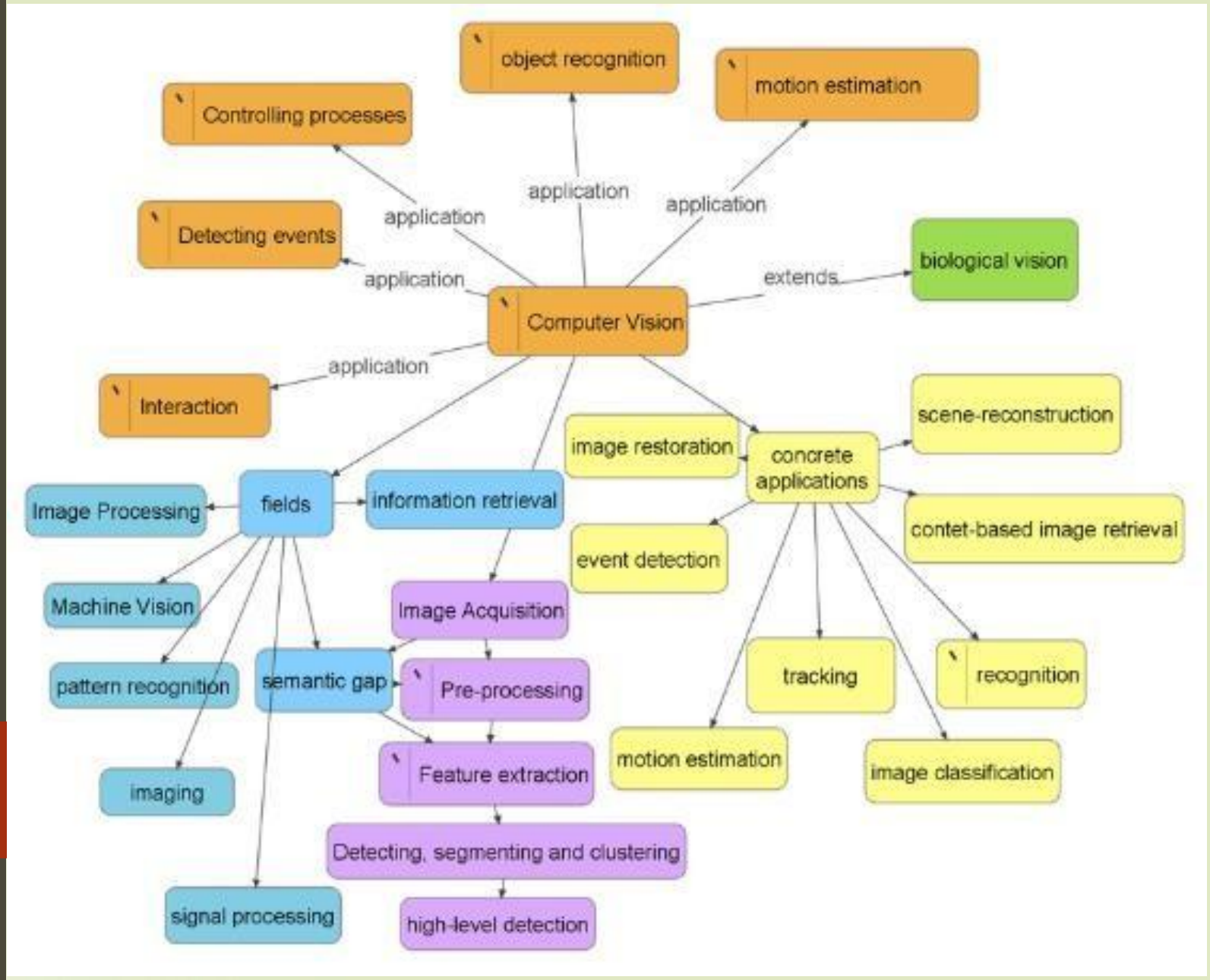

Probability Map of Soil Class


Probability Map of Soil Class


Probability Map of Soil Class


ROC, Auc(s) =0.92(Scenario 1),0.67(Scenario 2),0.73(Scenario 3)
Probability of Correct Detection
False Alarm Rate
Scenario 1
Scenario 2
Scenario 3

APPLICATIONS OF COMPUTER VISION – BROAD SCOPE



Applications of Computer Vision

- Face Recognition
- Object Recognition
- Video Surveillance and Monitoring
  - Object detection, tracking and behavior analysis
- Remote Sensing: UAVs
- Robotics
- Computer Graphics

Object Recognition



Finding People in images

Problem 1: Given an image I

Question: Does I contain an image of a person?

"Yes" Instances













"No" Instances













Localize People (Human Detection)



Human Detection





Individuals within small groups of people

Face Recognition



FACIAL EXPRESSIONS



RAISE EYE BROWS



SMILE

Detecting Driver Alertness





Video Surveillance and Monitoring



Object detection → Object tracking → Object categorization and classification → Event or Activities Recognition

- Automated Surveillance System (Detection & Tracking)







- SUMMARY:

- Scope of computer vision
- Historical perspective
- Research in Computer Vision
- Applications

